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Preventive and therapeutic effects of semiconductor laser on pain in root canal treatment

Ye Jian[1*]

Abstract

Objective This study aimed to evaluate the preventive and therapeutic effects of semiconductor laser on postoperative pain in root canal treatment and to compare the clinical efficacy of different laser application methods.

Methods A total of 202 patients requiring root canal treatment at our hospital's dental department from January to December 2024 were selected and randomly divided into experimental and Control groups using a random number table. The Control group received conventional root canal treatment (58 cases); the experimental group received 810 nm semiconductor laser-assisted treatment based on conventional root canal therapy and was further divided into three subgroups according to different laser application methods: laser-activated irrigation group (LAI group, 48 cases), low-level laser therapy group (LLLT group, 48 cases), and combined application group (Combined group, 48 cases). The LAI group used laser as a root canal irrigation activation system; the LLLT group applied laser irradiation to the buccal and palatal mucosa externally; the Combined group applied both methods simultaneously. All patients received root canal treatment completed in a single visit, and 197 patients completed the 14-day follow-up. Visual Analogue Scale (VAS) was used to assess the patients' pain levels at 1, 2, 3, 7, and 14 days after the procedure, and the use of analgesics was recorded.

Results Within 14 days after treatment, there were statistically significant differences in the mean VAS scores among the four groups ($P < 0.05$). The Control group had the highest pain scores, while the Combined group had the lowest. On the first day after treatment, the VAS scores of all experimental subgroups were significantly lower than the Control group ($P < 0.001$), with the LLLT group showing the best immediate pain relief effect. On days 2–3 after treatment, there was no significant difference in pain scores between the LAI and LLLT groups, but both were better than the Control group ($P < 0.001$). On day 7 after treatment, the Combined group had the most long-lasting analgesic effect, showing a significant difference compared to the Control group. By day 14, the differences among groups were no longer statistically significant ($P > 0.05$). Regarding analgesic use, the experimental groups required significantly less medication than the Control group ($P < 0.001$), with the Combined group using the least (0.53 ± 0.74 tablets). Moreover, multivariate analysis found that the size of periapical lesions, preoperative pain level, and laser treatment protocol were the main factors affecting postoperative pain.

Conclusion Semiconductor laser can effectively reduce postoperative pain after root canal treatment and improve patient comfort. The combined application of laser-activated irrigation and low-level laser therapy techniques is the optimal pain prevention and control protocol, which can continuously provide analgesic effects for 14 days after treatment. As an adjunctive treatment, semiconductor laser effectively reduces discomfort in root canal

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treatment by enhancing root canal disinfection and promoting inflammatory tissue repair. This study provides scientific evidence for the rational application of semiconductor laser in clinical root canal treatment, which is of great significance for improving the success rate and patient satisfaction of root canal treatment.

Keywords Semiconductor laser, Root canal treatment, Postoperative pain, Laser-activated irrigation, Low-level laser therapy, Preventive and therapeutic effects

Introduction

Root canal treatment, as an important means of preserving natural teeth, plays an irreplaceable role in modern dentistry. However, postoperative pain remains one of the main factors affecting patient acceptance and treatment satisfaction. Studies show that the incidence of pain after root canal treatment varies significantly, generally peaking within 24 h after surgery, with approximately 40% of patients experiencing different degrees of pain [1, 2]. This not only affects the quality of life of patients but may also lead to fear and resistance to subsequent treatments. Preoperative pain intensity, gender, and tooth type have been identified as key predictors of postoperative pain [3]. Traditional pain management relies mainly on oral analgesics, but their efficacy varies among individuals and may be accompanied by systemic side effects including gastrointestinal and renal complications [4]. Research indicates that non-steroidal anti-inflammatory drugs (especially when combined with acetaminophen), laser therapy, cryotherapy, and specific intracanal drug combinations can effectively alleviate postoperative pain [5]. Therefore, developing more effective and safer postoperative pain prevention and treatment strategies based on this new evidence has become a focus of current endodontic research.

In recent years, the application of laser technology in dentistry has become increasingly widespread, with its role in root canal treatment gaining attention. Semiconductor lasers, due to their excellent tissue penetration, precise energy control, and good biocompatibility, demonstrate unique advantages in root canal treatment [6, 7]. Studies show that laser-assisted root canal treatment can not only improve disinfection effects but also reduce postoperative pain by modulating inflammatory responses [8, 9].

The application of lasers in root canal treatment primarily includes two methods: Laser-Activated Irrigation (LAI) and Low-Level Laser Therapy (LLLT). LAI uses erbium laser pulses to act on the irrigation solution in the root canal, producing cavitation effects and acoustic streaming phenomena, enhancing the cleaning and disinfection of the root canal system [10, 11]. Systematic reviews indicate that Er:YAG LAI can significantly reduce postoperative pain intensity within 48 h after root canal treatment, especially when using Photon-Induced Photoacoustic Streaming (PIPS) technology [12]. LLLT, on the other hand, promotes tissue repair and reduces pain through photobiological regulation, affecting mitochondrial activity and inflammatory factors in cells [13]. Multiple randomized controlled clinical trials have demonstrated that treatment with 980 nm wavelength semiconductor lasers can significantly reduce postoperative pain after root canal retreatment, particularly showing significant advantages in pain control within 4–48 h after surgery [14].

Currently, there are still some deficiencies in clinical research on semiconductor lasers in the prevention and treatment of postoperative pain in root canal treatment. First, most studies have relatively small sample sizes, lacking sufficient statistical power; second, the standardization of research designs is not high, making effective comparison and analysis difficult; third, few studies simultaneously compare the differences in effects between different laser application methods. In addition, the optimization of laser treatment parameters, determination of indications, and in-depth understanding of treatment mechanisms still require more high-quality clinical studies to provide evidence support.

This study designed a prospective randomized controlled clinical trial aimed at systematically evaluating the clinical efficacy of 810 nm semiconductor laser in preventing and treating postoperative pain in root canal treatment, and comparing the differences in efficacy among LAI, LLLT, and their combined application. Through multidimensional efficacy evaluation and in-depth factor analysis, this study hopes to provide more reliable clinical evidence for the rational application of laser technology in root canal treatment and promote the development of pain-free treatment concepts in endodontics.

Materials and methods

Research design

This study was designed as a prospective randomized controlled clinical trial, approved by the Wuxi

Stomatological Hospital Medical Ethics Committee (Approval No.: WXKQ-2023-015), and all participating patients signed informed consent forms. The research followed the ethical principles of the Declaration of Helsinki.

Study population

Sample size calculation

Based on the results of a preliminary trial, G*Power software (Version 3.1.9.7, Germany) was used to calculate the sample size. With $\alpha = 0.05$, $1 - \beta = 0.90$, and an effect size of 0.25, analysis of variance estimation determined that at least 188 patients were needed. Considering a potential dropout rate of approximately 10% during follow-up, the final sample size was determined to be 210 patients, with 202 patients actually enrolled in this study.

Inclusion and exclusion criteria

Inclusion criteria (1) Age 18–65 years, gender not restricted; (2) Patients diagnosed with pulpitis or apical periodontitis requiring root canal treatment based on clinical symptoms, electric pulp testing, and radiographic examination; (3) Single or multi-rooted teeth with basic periodontal health (probing depth ≤ 3 mm); (4) No history of systemic diseases or special drug allergies; (5) No consumption of medications affecting pain perception (such as analgesics, sedatives, etc.) within the past month; (6) Willing to cooperate with treatment and follow-up, and able to correctly understand and complete the pain assessment form.

Exclusion criteria (1) Pregnant or lactating women; (2) Patients with severe systemic diseases, such as uncontrolled diabetes, cardiovascular disease, etc.; (3) Long-term use of medications affecting pain perception; (4) Teeth with vertical root fractures, severe periodontal lesions (probing depth ≥ 5 mm), or severe mobility (mobility ≥ Grade II); (5) Teeth with acute periapical abscess or severe periapical lesions (diameter ≥ 10 mm); (6) Patients with mental disorders or communication barriers; (7) Cases where severe complications during treatment necessitated a change in treatment plan.

To address potential confounding factors related to tooth type diversity, stratified randomization was employed to ensure balanced distribution of single-rooted and multi-rooted teeth across all treatment groups. Although tooth type is a known predictor of postoperative pain, the balanced distribution ensures comparability between groups, and multivariate analysis was performed to control for this factor.

Patients with pulpal necrosis were diagnosed based on negative response to electric pulp testing, absence of bleeding upon pulp chamber access, and clinical symptoms consistent with necrotic pulp tissue.

Grouping and randomization

Patients were divided into experimental and Control groups using a computer-generated random number table (SPSS 26.0, IBM, USA). The Control group consisted of 58 cases, while the experimental group was divided into three subgroups according to laser application method: laser-activated irrigation group (LAI group, 48 cases), low-level laser therapy group (LLLT group, 48 cases), and combined application group (Combined group, 48 cases). The randomization sequence was generated by a statistical professional not involved in clinical treatment, and allocation concealment was implemented using the sealed opaque envelope method. The treating operator could only open the envelope to obtain group information after completing the conventional root canal preparation.

Treatment protocols

Preoperative assessment

All patients underwent detailed preoperative examinations: (1) Chief complaint and medical history collection; (2) Clinical examination: visual inspection, percussion, palpation, mobility assessment, periodontal probing, etc.; (3) Radiographic examination: periapical radiographs to evaluate periapical conditions; (4) Pain assessment: using the Visual Analogue Scale (VAS, 0–10 points, 0 points indicating no pain, 10 points indicating unbearable severe pain) to record the patient's preoperative pain level.

Standard root canal treatment procedure

All patients received treatment from the same dentist with over 10 years of clinical experience, following this protocol:

(1) Local anesthesia: 2% lidocaine with 1:100,000 epinephrine (Huarun Double-Crane Pharmaceutical, China) for local infiltration or regional block anesthesia; (2) Rubber dam isolation: using a rubber dam (Hygenic, Coltene/Whaledent Inc., USA) to isolate the operation area and create a sterile environment; (3) Access creation: using a high-speed turbine to access the pulp chamber, expose the pulp cavity, and locate the canal orifices; (4) Working length determination: using an electronic apex locator (Root ZX II, J. Morita, Japan) combined with radiographic verification to determine the working length, set as 0.5 mm short of the apical constriction; (5) Root canal preparation: using the ProTaper Next system (Dentsply Sirona, Switzerland) for mechanical preparation of the root canal, working length to X3 (ISO 30) or X4 (ISO 40),

depending on the initial canal diameter; (6) Root canal irrigation: using 5.25% sodium hypochlorite solution (20 ml, ultrasonic agitation 20 s × 3 times) and 17% EDTA solution (5 ml, ultrasonic agitation 60 s × 1 time) alternately, with final irrigation using sterile saline solution (10 ml); (7) Root canal drying: using paper points (Dentsply Sirona, Switzerland) matching the final preparation size to dry the canal; (8) Root canal filling: employing continuous wave vertical compaction technique (Elements Free, Kerr, USA) with AH Plus root canal sealer (Dentsply Sirona, Switzerland) and gutta-percha points (Dentsply Sirona, Switzerland); (9) Postoperative restoration: using light-cured composite resin (Filtek Z350 XT, 3 M ESPE, USA) for permanent coronal filling.

Laser-assisted treatment protocols

After completing root canal preparation and conventional irrigation, different laser-assisted treatments were administered to the experimental group patients according to their group allocation:

LAI group (Laser-Activated Irrigation Group)

Using an 810 nm semiconductor laser device (Biolase, USA) with a 200-μm-diameter optical fiber tip, parameters were set as follows: power 1.5W, pulse mode (50 Hz, 50% duty cycle), and effective power 0.75W. The 810 nm wavelength was selected for LAI based on its excellent tissue penetration properties and ability to generate photoacoustic effects in irrigation solutions, while providing the additional benefit of allowing the same laser system to be used for both LAI and LLLT applications in this study. The root canal was filled with 5.25% sodium hypochlorite solution, and the laser fiber was inserted to a point 1 mm short of the working length, moving in a spiral upward motion. Each canal was activated for 20 s, repeated 3 times, with 10-s intervals between each activation, continuously replenishing the irrigation solution during activation.

LLLT group (Low-Level Laser Therapy Group)

Using the same 810 nm semiconductor laser device, with an 8-mm-diameter flat probe, parameters were set as follows: power 0.5W, continuous mode, and energy density 4 J/cm^2. Irradiation was applied to the buccal and palatal/lingual mucosa at the corresponding tooth site for 60 s each, maintaining the probe 1-2 mm from the tissue surface, perpendicular to the tissue surface. The buccal and palatal/lingual mucosa irradiation sites were selected based on previous studies demonstrating that transcutaneous LLLT at 810 nm wavelength can effectively penetrate tissues to reach periapical areas with penetration depths of 2-4 cm, while avoiding direct intraoral manipulation that could compromise the sterile operative field.

Combined group (combined application group)

LAI treatment was performed first, followed by LLLT treatment, with the operating parameters for both methods remaining the same as when used individually.

Control group

After completing conventional root canal treatment, a simulated operation was performed using the laser device with the power turned off (placebo effect control), with the operating time being the same as the experimental group.

All patients completed their root canal treatment in a single visit. The operation time was recorded during treatment, and patients were allowed to rest for 15 min after treatment and were assessed for immediate reactions before discharge.

Postoperative assessment

Postoperative pain assessment

Patients were provided with standardized VAS scoring forms and pain diaries, required to record pain conditions at 1, 2, 3, 7, and 14 days after treatment. Pain levels were categorized as follows: no pain (0 points), mild pain (1–3 points), moderate pain (4–6 points), and severe pain (7–10 points). Data were collected through a combination of telephone follow-up and return visits.

Analgesic medication usage

Patients were uniformly provided with ibuprofen sustained-release capsules (300 mg/tablet, Xiuzheng Pharmaceutical, China) and instructed to take one tablet for moderate pain (VAS ≥ 4 points), with a maximum of 3 tablets per day, at intervals of not less than 8 h. Patients were advised to discontinue analgesic use once pain subsided to VAS ≤ 3 points, with medication duration typically not exceeding 3–5 days to minimize interference with pain assessment. Patients were required to record the time and quantity of medication in detail.

Assessment of clinical signs and symptoms

At the 14-day follow-up visit, patients' clinical signs and symptoms were evaluated: percussion response, occlusal pain, spontaneous pain, gingival swelling, and sinus tract formation.

Data collection and management

All data were independently collected by two trained research assistants who did not participate in patient treatment and were blinded to the grouping

information. Data were entered into pre-coded electronic spreadsheets, with duplicate databases established and cross-checked to ensure accuracy. Multiple imputation methods were used to handle missing data.

Statistical analysis

SPSS 26.0 software (IBM, USA) was used for statistical analysis. Quantitative data are expressed as mean ± standard deviation ($\bar{x} \pm s$), and qualitative data as frequencies and percentages. The following statistical methods were used for between-group comparisons:(1) Comparison of baseline data between groups: quantitative data using one-way analysis of variance or Kruskal–Wallis test; qualitative data using chi-square test or Fisher's exact test; (2) Comparison of pain scores between groups: using Kruskal–Wallis test and Mann–Whitney U test, with Bonferroni correction for multiple comparisons; (3) Comparison of pain scores at different time points within groups: using Friedman test and Wilcoxon signed-rank test; (4) Comparison of analgesic medication usage: using one-way analysis of variance or Kruskal–Wallis test; (5) Multivariate logistic regression analysis to identify factors influencing postoperative pain.All tests were two sided, with $P < 0.05$ considered statistically significant.

Results

Study flow and baseline characteristics

This study recruited 210 patients who met the inclusion criteria, of whom 202 were randomly assigned to four treatment groups after screening: LAI group (48 cases), LLLT group (48 cases), Combined group (48 cases), and Control group (58 cases). During the 14-day follow-up period, a total of 5 patients were lost to follow-up (dropout rate 2.5%), with 197 patients ultimately completing the entire study.

The baseline demographic and clinical characteristics were balanced across the four groups, with no statistical differences in age, gender, tooth position distribution, pulp status, periapical diagnosis, and other aspects ($P > 0.05$) (see Table 1). Preoperative pain scores, periapical lesion size, and treatment duration also showed no significant differences between groups, ensuring intergroup comparability.

Postoperative pain scores

The VAS pain scores of patients in all groups showed a time-dependent decreasing trend (see Fig. 1). On the first day after treatment, there were significant differences in pain scores among the four groups ($P < 0.001$), with the Combined group having the lowest pain score (2.45 ± 1.74), significantly lower than the LAI group (3.21 ± 1.85), LLLT group (2.89 ± 1.90), and Control group (4.58 ± 2.10).

On the second and third days after treatment, the pain scores of the three experimental groups were all significantly lower than the Control group ($P < 0.001$), with the Combined group showing the best analgesic effect, demonstrating statistical differences compared to both the LAI group and LLLT group ($P < 0.05$). By the seventh day after treatment, only the Combined group and Control group still showed significant differences ($P < 0.05$). By

Table 1 Baseline demographic and clinical characteristics of patients

Group	Control ($n = 58$)	LAI ($n = 48$)	LLLT ($n = 48$)	Combined ($n = 48$)	*P*-value
Age (years)	41.5 ± 11.2	42.8 ± 10.5	40.9 ± 12.1	42.3 ± 10.8	0.768
Gender (Male/Female)	27/31	23/25	22/26	23/25	0.891
Tooth type					
Anterior	19 (32.8%)	16 (33.3%)	15 (31.3%)	16 (33.3%)	0.992
Premolar	23 (39.7%)	19 (39.6%)	19 (39.6%)	19 (39.6%)	0.999
Molar	16 (27.6%)	13 (27.1%)	14 (29.2%)	13 (27.1%)	0.984
Pulpal diagnosis					
Irreversible pulpitis	29 (50.0%)	24 (50.0%)	23 (47.9%)	24 (50.0%)	0.979
Pulp necrosis	23 (39.7%)	19 (39.6%)	20 (41.7%)	19 (39.6%)	0.987
Previously treated	6 (10.3%)	5 (10.4%)	5 (10.4%)	5 (10.4%)	0.999
Periapical diagnosis					
Normal	20 (34.5%)	16 (33.3%)	15 (31.3%)	17 (35.4%)	0.934
Symptomatic apical periodontitis	26 (44.8%)	22 (45.8%)	22 (45.8%)	21 (43.8%)	0.986
Asymptomatic apical periodontitis	12 (20.7%)	10 (20.8%)	11 (22.9%)	10 (20.8%)	0.979
Size of periapical lesion (mm)	2.7 ± 2.1	2.4 ± 1.9	2.6 ± 2.2	2.5 ± 2.0	0.901
Preoperative pain (VAS score)	3.3 ± 2.6	3.4 ± 2.4	3.2 ± 2.5	3.3 ± 2.6	0.935
Treatment duration (min)	52.1 ± 13.5	54.8 ± 12.7	53.7 ± 14.1	57.2 ± 13.3	0.241

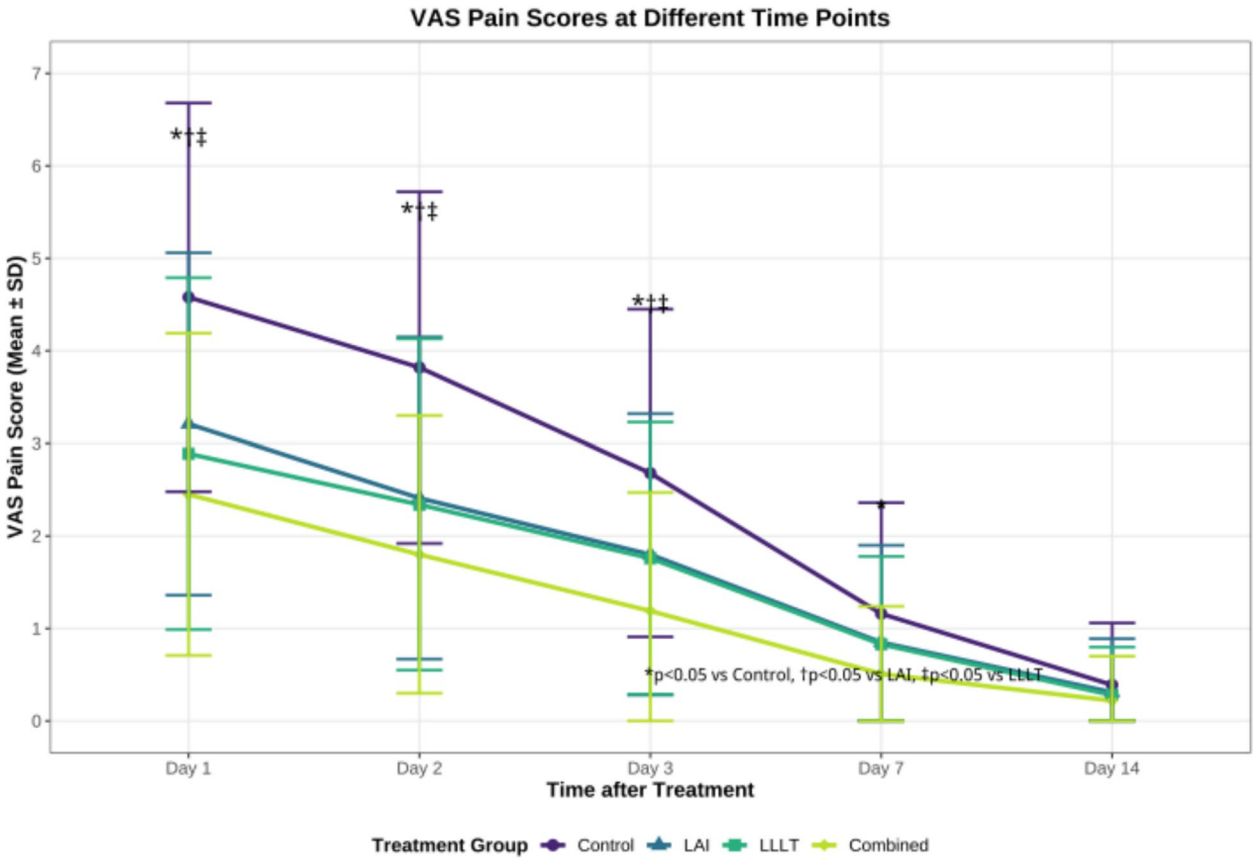


Fig. 1 Comparison of VAS pain scores among different treatment groups over time

Table 2 Mean VAS pain scores at different time points

Group	Day 1	Day 2	Day 3	Day 7	Day 14
Control ($n=56$)	4.58 ± 2.10	3.82 ± 1.90	2.68 ± 1.77	1.16 ± 1.20	0.39 ± 0.67
LAI ($n=47$)	3.21 ± 1.85*	2.41 ± 1.74*	1.80 ± 1.52*	0.85 ± 1.05	0.31 ± 0.58
LLLT ($n=47$)	2.89 ± 1.90*	2.34 ± 1.79*	1.76 ± 1.47*	0.83 ± 0.95	0.28 ± 0.52
Combined ($n=47$)	2.45 ± 1.74*†‡	1.80 ± 1.50*†‡	1.19 ± 1.28*†‡	0.51 ± 0.73*	0.22 ± 0.48
P-value	< 0.001	< 0.001	< 0.001	0.008	0.438

Values are presented as mean ± standard deviation. *$P < 0.05$ compared with Control group; †$P < 0.05$ compared with LAI group; ‡$P < 0.05$ compared with LLLT group

the 14th day after treatment, there were no longer any statistical differences in pain scores between the groups ($P=0.438$) (see Table 2).

Regarding the distribution of pain intensity (see Table 3; Fig. 2), the proportion of patients with moderate to severe pain (VAS ≥ 4) on the first day after treatment was significantly higher in the Control group (62.5%) than in the three experimental groups (LAI group 40.4%, LLLT group 34.1%, Combined group 21.2%) ($P<0.001$). Throughout the follow-up period, the proportion of patients with no pain or mild pain in the experimental groups was consistently higher than in the Control group, with this advantage being particularly evident during the first three days after treatment.

Analgesic medication usage

In terms of analgesic medication usage (see Table 4; Fig. 3), 75.0% of patients in the Control group required analgesics during the follow-up period, significantly higher than the LAI group (51.1%), LLLT group (48.9%), and Combined group (36.2%) ($P<0.001$). The average analgesic consumption per patient was highest

Table 3 Distribution of pain intensity at different time points

Time point	Pain intensity	Control ($n=56$)	LAI ($n=47$)	LLLT ($n=47$)	Combined ($n=47$)	*P*-value
Day 1	No pain (0)	5 (8.9%)	7 (14.9%)	8 (17.0%)	11 (23.4%)	< 0.001
	Mild (1–3)	16 (28.6%)	21 (44.7%)	23 (48.9%)	26 (55.3%)	
	Moderate (4–6)	26 (46.4%)	16 (34.0%)	13 (27.7%)	9 (19.1%)	
	Severe (7–10)	9 (16.1%)	3 (6.4%)	3 (6.4%)	1 (2.1%)	
Day 2	No pain (0)	7 (12.5%)	11 (23.4%)	13 (27.7%)	18 (38.3%)	< 0.001
	Mild (1–3)	20 (35.7%)	24 (51.1%)	22 (46.8%)	22 (46.8%)	
	Moderate (4–6)	22 (39.3%)	10 (21.3%)	10 (21.3%)	7 (14.9%)	
	Severe (7–10)	7 (12.5%)	2 (4.3%)	2 (4.3%)	0 (0.0%)	
Day 3	No pain (0)	13 (23.2%)	17 (36.2%)	18 (38.3%)	24 (51.1%)	0.003
	Mild (1–3)	26 (46.4%)	24 (51.1%)	22 (46.8%)	21 (44.7%)	
	Moderate (4–6)	15 (26.8%)	6 (12.8%)	7 (14.9%)	2 (4.3%)	
	Severe (7–10)	2 (3.6%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	
Day 7	No pain (0)	28 (50.0%)	27 (57.4%)	28 (59.6%)	33 (70.2%)	0.035
	Mild (1–3)	25 (44.6%)	18 (38.3%)	17 (36.2%)	14 (29.8%)	
	Moderate (4–6)	3 (5.4%)	2 (4.3%)	2 (4.3%)	0 (0.0%)	
	Severe (7–10)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	
Day 14	No pain (0)	45 (80.4%)	38 (80.9%)	40 (85.1%)	42 (89.4%)	0.641
	Mild (1–3)	11 (19.6%)	9 (19.1%)	7 (14.9%)	5 (10.6%)	
	Moderate (4–6)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	
	Severe (7–10)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	

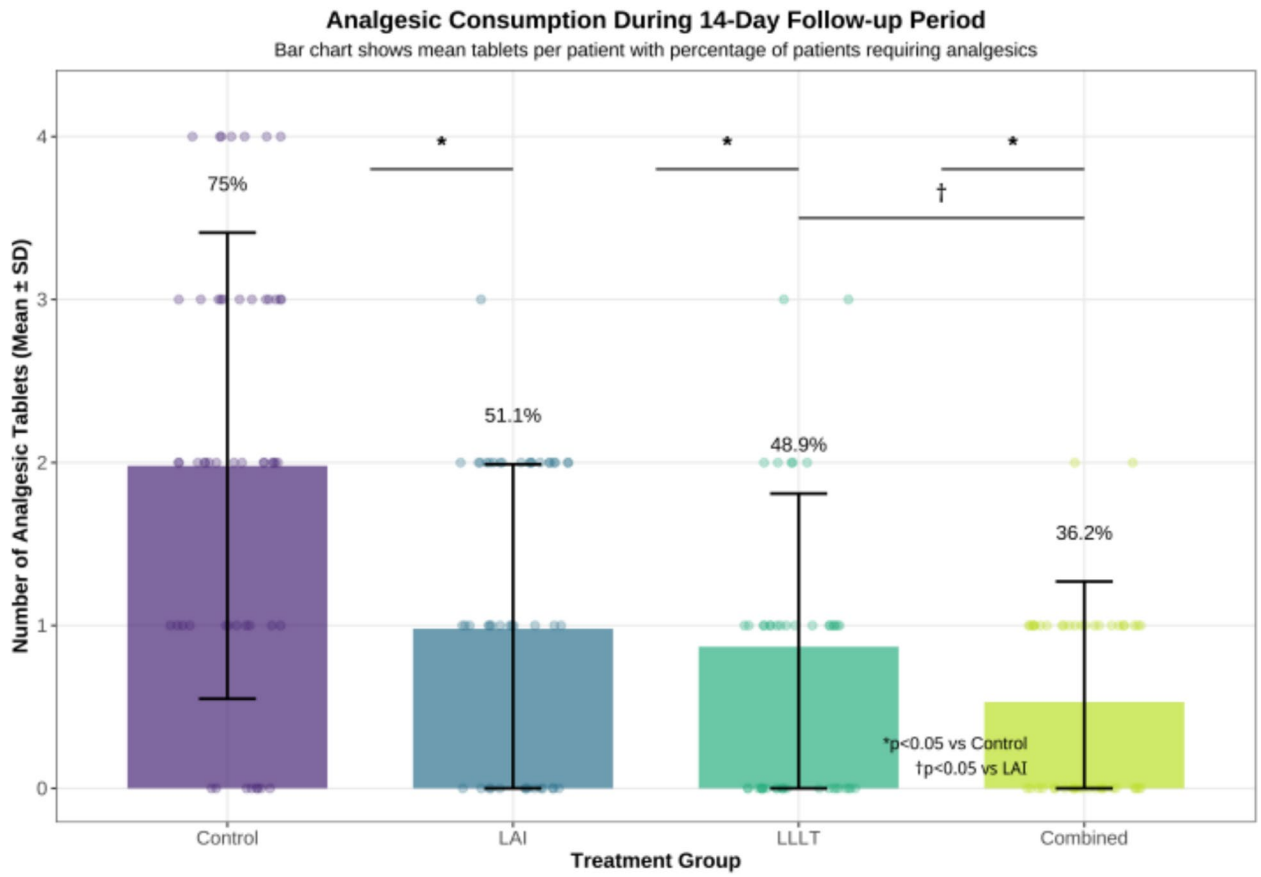


Fig. 2 Distribution of pain intensity categories at early postoperative time points by treatment group

Table 4 Analgesic consumption during the follow-up period

Group	Patients requiring analgesics n (%)	Total number of analgesic tablets	Mean tablets per patient
Control ($n=56$)	42 (75.0%)	111	1.98 ± 1.43
LAI ($n=47$)	24 (51.1%)*	46	0.98 ± 1.01*
LLLT ($n=47$)	23 (48.9%)*	41	0.87 ± 0.94*
Combined ($n=47$)	17 (36.2%)*	25	0.53 ± 0.74*†
P-value	< 0.001	< 0.001	< 0.001

Values are presented as number (percentage) or mean ± standard deviation.
*$P<0.05$ compared with Control group; †$P<0.05$ compared with LAI group

in the Control group (1.98 ± 1.43 tablets) and lowest in the Combined group (0.53 ± 0.74 tablets), with a statistically significant difference compared to the LAI group (0.98 ± 1.01 tablets) (P < 0.05), but no significant difference compared to the LLLT group (0.87 ± 0.94 tablets).

Analysis of factors influencing postoperative pain

The results of multivariate logistic regression analysis (see Table 5) showed that the treatment protocol was an important factor influencing moderate to severe postoperative pain (VAS ≥ 4). Compared with the Control group, the Combined group had the lowest risk of moderate to severe postoperative pain (OR = 0.15, 95%CI: 0.06–0.39, $P<0.001$), followed by the LLLT group (OR = 0.29, 95%CI: 0.13–0.68, $P=0.004$) and the LAI group (OR = 0.38, 95%CI: 0.17–0.86, $P=0.020$).

Moderate preoperative pain (OR = 2.34, 95%CI: 1.15–4.76, $P=0.019$) and severe preoperative pain (OR = 3.87, 95%CI: 1.46–10.25, $P=0.006$) were risk factors for postoperative pain. Additionally, patients with periapical lesion diameters > 5 mm had a significantly increased risk of moderate to severe postoperative pain (OR = 2.91, 95%CI: 1.18–7.16, $P=0.020$). Age, gender, tooth position, and pulp diagnosis type showed no significant correlation with postoperative pain.

Adverse reactions

No significant adverse reactions related to laser treatment were observed in any of the experimental group patients. Two patients (1 in the LAI group, 1 in the Combined group) reported a mild warm sensation during laser irradiation, but both could tolerate it without interruption of treatment. None of the patients experienced serious complications related to root canal treatment during the follow-up period, such as sinus tract formation, persistent swelling, or severe infection.

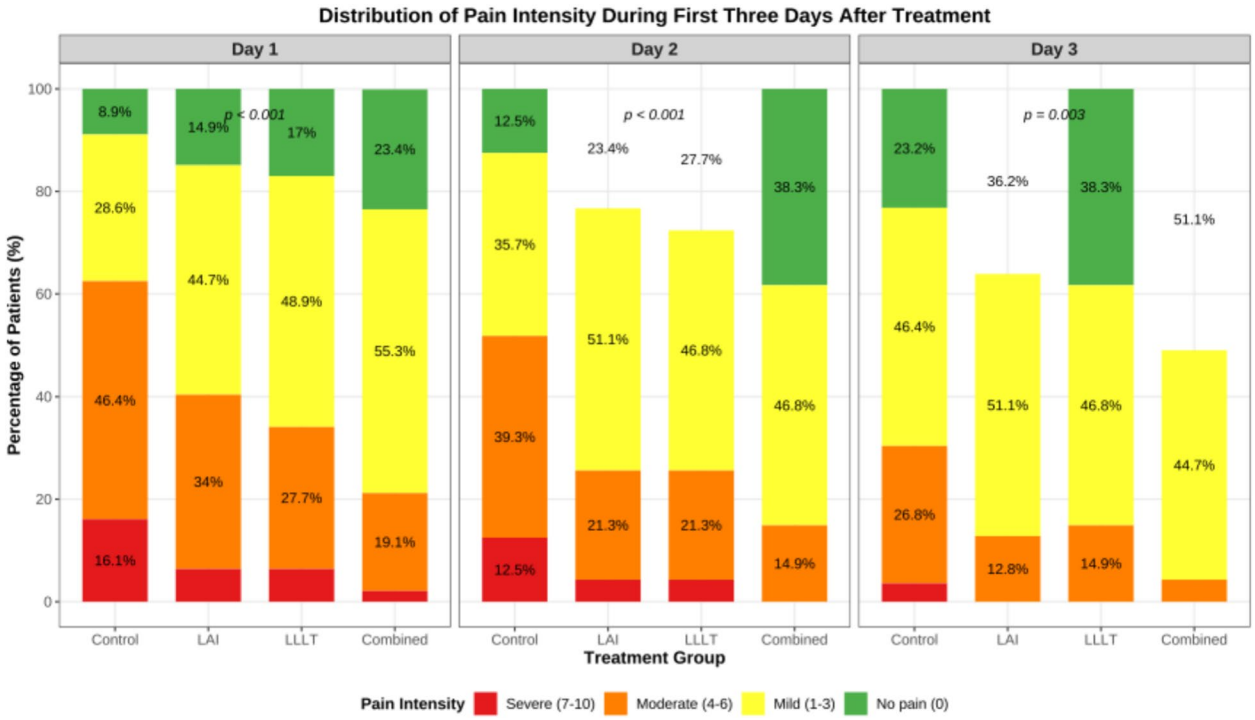


Fig. 3 Analgesic consumption and usage patterns among treatment groups during 14-day follow-up period

Table 5 Multivariable logistic regression analysis for postoperative moderate to severe pain (VAS ≥ 4)

Factor	Odds ratio	95% Confidence Interval	*P*-value
Treatment group			
Control	Reference		
LAI	0.38	0.17–0.86	0.020
LLLT	0.29	0.13–0.68	0.004
Combined	0.15	0.06–0.39	< 0.001
Preoperative pain (VAS score)			
0–3 (Mild)	Reference		
4–6 (Moderate)	2.34	1.15–4.76	0.019
7–10 (Severe)	3.87	1.46–10.25	0.006
Size of periapical lesion			
< 2 mm	Reference		
2–5 mm	1.72	0.82–3.59	0.149
> 5 mm	2.91	1.18–7.16	0.020
Pulpal diagnosis			
Irreversible pulpitis	Reference		
Pulp necrosis	1.31	0.64–2.66	0.457
Previously treated	1.89	0.73–4.89	0.189
Age	1.01	0.98–1.04	0.512
Gender			
Male	Reference		
Female	1.23	0.66–2.30	0.518
Tooth type			
Anterior	Reference		
Premolar	1.14	0.53–2.44	0.741
Molar	1.68	0.76–3.74	0.203

Discussion

This study evaluated the clinical efficacy of an 810 nm semiconductor laser in preventing and treating postoperative pain in root canal therapy and compared the therapeutic effects of different laser application methods. The results demonstrated that semiconductor lasers can alleviate postoperative pain following root canal treatment, with the combined application of laser-activated irrigation (LAI) and low-level laser therapy (LLLT) showing the most optimal analgesic effect.

Postoperative pain is one of the most common complications of root canal treatment, with an incidence rate of 25–40% [15, 16]. Traditional pain management primarily relies on oral analgesics, which have certain limitations, including significant individual variations, potential gastrointestinal adverse reactions, and inadequate efficacy for some patients [17]. This study confirms that semiconductor lasers, as a non-pharmacological intervention, can effectively reduce postoperative pain intensity and decrease analgesic medication usage without increasing systemic side effects.

The treatment mechanism of the 810 nm wavelength semiconductor laser used in this study involves multiple aspects. First, regarding laser-activated irrigation (LAI), laser energy produces cavitation phenomena through photoacoustic effects, enhancing the flow and mixing of irrigants and improving the efficiency of microbial elimination in the root canal [18, 19]. The cavitation effects induced by laser activation generate powerful fluid dynamic effects within the root canal, effectively removing bacterial biofilms and debris [20]. Additionally, the photothermal effects of lasers can mitigate inflammatory responses and promote tissue repair, thereby reducing postoperative pain [21]. Recent systematic reviews indicate that laser-assisted treatment can significantly reduce pain levels within 48 h after root canal treatment [22], which is consistent with our findings.

Regarding low-level laser therapy (LLLT), the 810 nm wavelength is within the near-infrared spectrum range and has good tissue penetration. The transcutaneous application approach utilized in this study was based on the principle that 810 nm laser light can effectively penetrate oral tissues to reach periapical areas, providing therapeutic benefits while maintaining the sterile operative environment. This method has been validated in previous studies showing effective tissue penetration depths of 2-4 cm for 810 nm wavelength, which is sufficient to reach most periapical regions from the buccal and lingual aspects. The therapeutic mechanism of LLLT is primarily based on photobiomodulation, with the core being the promotion of cellular energy metabolism through activation of mitochondrial cytochrome c oxidase (CCO) [23, 24]. Research shows that when 810 nm laser irradiates tissues, it can enhance mitochondrial membrane potential, promote ATP synthesis, and reach peak effects 3–6 h after irradiation [25]. Furthermore, LLLT can regulate the production of reactive oxygen species (ROS) and nitric oxide (NO), thereby influencing the activity of various transcription factors, including NF-κB [26]. Clinical studies confirm that LLLT can downregulate pro-inflammatory factors such as TNF-α and IL-1β, while upregulating anti-inflammatory factors such as IL-10 [27]. These molecular mechanisms collectively form the foundation for its analgesic effects.

The innovation of this study lies in the simultaneous comparison of three different laser application methods: LAI, LLLT, and their combined application. The results show that while the application of either LAI or LLLT alone can significantly reduce postoperative pain, the combined application protocol demonstrated the most significant effect, suggesting a potential synergistic action between the two laser treatment modalities. LAI primarily prevents postoperative pain by improving

root canal disinfection effects, with its mechanism focusing on eliminating pathogenic bacteria and endotoxins, thereby reducing infection sources [28]. LLLT, on the other hand, primarily alleviates pain by promoting mitochondrial ATP generation, regulating inflammatory responses, and facilitating tissue repair [29]. The combination of these approaches not only enables more thorough elimination of infection sources within the root canal but also promotes the repair of periapical tissues through biological regulation.

Through multivariate analysis, this study identified the main factors influencing postoperative pain, including treatment protocol, preoperative pain level, and periapical lesion size. Preoperative pain level was an important predictor of postoperative pain, which is consistent with the findings of previous studies [30]. Patients with moderate to severe preoperative pain may already have more severe inflammatory responses and neural sensitization, making them more susceptible to postoperative pain. Preventive application of laser therapy may have greater clinical value for these high-risk patients.

Although this study has significant clinical value, there are several limitations that should be considered when interpreting the results: First, this was a single-center study conducted at our institution, which may introduce selection bias and potentially limit the generalizability of the results to other clinical settings [31, 32]. Multi-center studies would help validate the universality of the findings across different populations and treatment protocols. Second, while the 14-day follow-up period adequately captured the acute pain phase when most clinically significant postoperative pain occurs, a longer follow-up period would be valuable to assess the durability of laser therapy benefits and potential effects on long-term healing outcomes, including periapical tissue healing and overall root canal treatment success rates [33, 34]. Third, the as-needed analgesic protocol, while reflecting real clinical practice and patient autonomy, may have partially masked the true magnitude of the analgesic effect of laser therapy. However, the significantly reduced analgesic consumption in laser groups compared to controls suggests that this factor did not substantially compromise the validity of our primary outcomes. Fourth, due to the special nature of laser equipment, a true double-blind design could not be achieved, potentially introducing a certain risk of bias. Furthermore, although baseline characteristics were similar across groups, there remains some patient heterogeneity, including the inclusion of both single-rooted and multi-rooted teeth as well as patients with varying pulpal diagnoses, that may affect the assessment of treatment efficacy.

Regarding long-term clinical outcomes, while this study focused primarily on short-term postoperative pain management, the improved root canal disinfection achieved through LAI and the anti-inflammatory effects of LLLT may potentially contribute to enhanced periapical healing and improved overall success rates of root canal therapy. Future studies with extended follow-up periods should investigate these potential long-term benefits, including radiographic assessment of periapical healing and clinical success rates at 6-month and 1-year intervals.

The clinical implications of this study suggest that semiconductor laser therapy, particularly the combined LAI and LLLT protocol, represents a valuable adjunctive treatment modality that can significantly improve patient comfort during the critical postoperative period. This non-pharmacological approach may be particularly beneficial for patients who cannot tolerate conventional analgesics or those seeking to minimize medication use.

Author contribution
YJ conceptulized the work and wrote the main manuscript text.

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Data availability
Data are available on request from author.

Declarations

Ethical approval
Ethical approval was not required for this study due to its retrospective/analytical nature, which complied with institutional and national ethical guidelines for non-interventional research.

Consent to participate
Patient consent was not required for this study as it did not involve human participants or personal data requiring informed approval.

Competing interest
The authors declare no competing interests.

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